# program 12

AUTHOR
kumkum mk 1nt23is106

## 12 Develop a script in R to create a violin plot displaying the distribution of a continuous variable , with a separate violins for each group using an inbuilt dataset.

## Step 1: Load the library

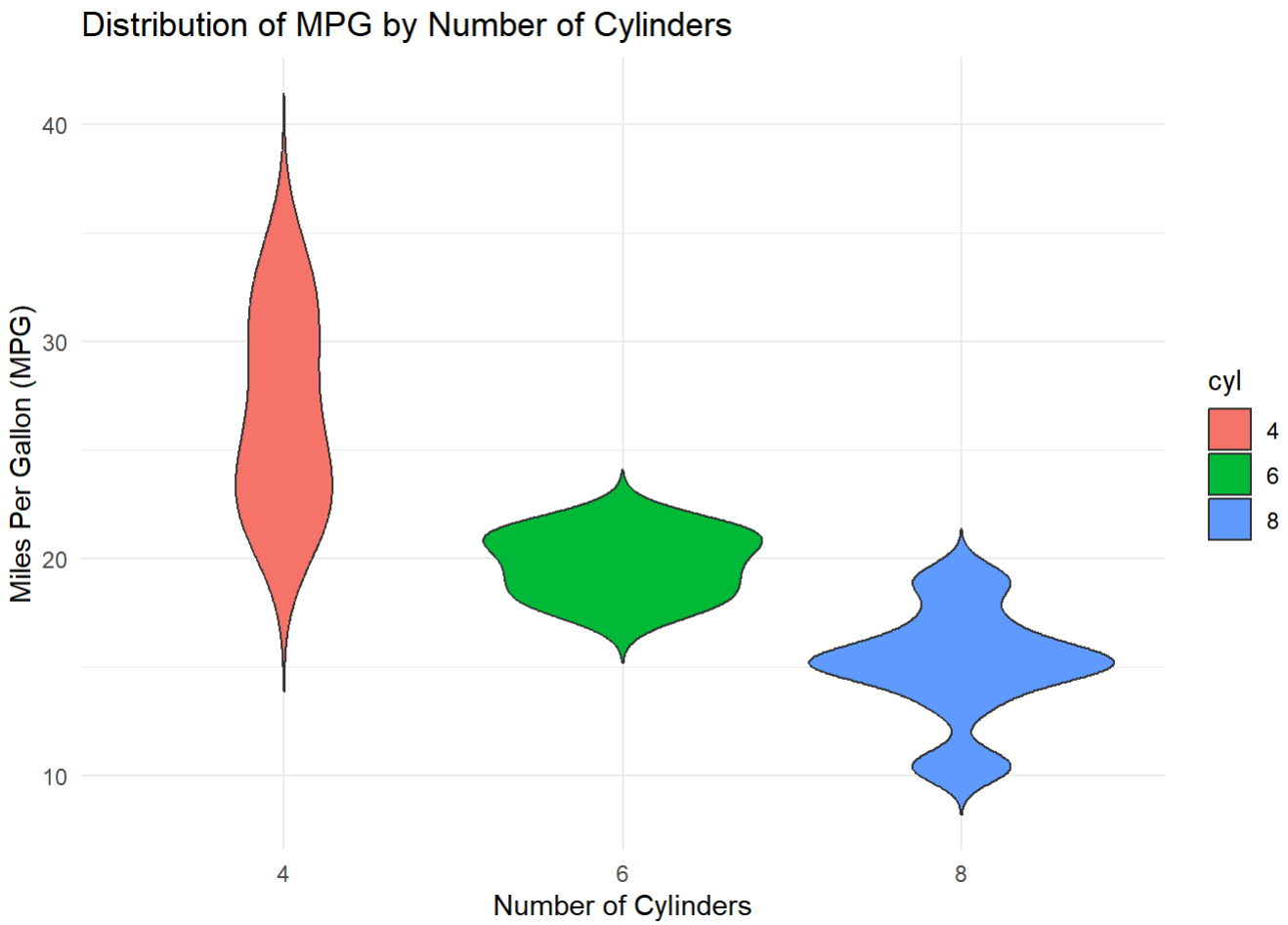
```
library(ggplot2)
```

## Step 2: Grouping variable

```
mtcars$cyl <- as.factor(mtcars$cyl)
```

## Step 3: Create a violin plot

```
ggplot(mtcars, aes(x = cyl, y = mpg, fill = cyl)) +
  geom_violin(trim = FALSE) +
  labs(
    title = "Distribution of MPG by Number of Cylinders",
    x = "Number of Cylinders",
    y = "Miles Per Gallon (MPG)"
  ) +
  theme_minimal()
```


Distribution of MPG by Number of Cylinders
Miles Per Gallon (MPG)
40
30
20
10
4
6
8
Number of Cylinders
cyl
4
6
8